SnpEff: Variant analysis

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Summary

Genome	GRCh38.99
Date	2025-11-14 21:44
SnpEff version	`SnpEff 5.3a (build 2025-09-02 10:24), by Pablo Cingolani`
Command line arguments	`SnpEff  GRCh38.99 raw.vcf`
Warnings	583,543
Errors	0
Number of lines (input file)	813,390
Number of variants (before filter)	815,355
Number of non-variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	814,996
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	1,965
Number of annotations	4,021,035
Genome total length	63,147,197,748
Genome effective length	3,099,061,249
Variant rate	1 variant every 3,802 bases

Variants rate details

Chromosome	Length	Variants	Variants rate
1	248,956,422	80,760	3,082
2	242,193,529	62,416	3,880
3	198,295,559	47,975	4,133
4	190,214,555	31,353	6,066
5	181,538,259	41,290	4,396
6	170,805,979	47,684	3,582
7	159,345,973	40,061	3,977
8	145,138,636	38,078	3,811
9	138,394,717	33,574	4,122
10	133,797,422	40,682	3,288
11	135,086,622	39,866	3,388
12	133,275,309	33,622	3,963
13	114,364,328	24,572	4,654
14	107,043,718	20,652	5,183
15	101,991,189	24,769	4,117
16	90,338,345	32,304	2,796
17	83,257,441	32,657	2,549
18	80,373,285	20,405	3,938
19	58,617,616	35,979	1,629
20	64,444,167	23,473	2,745
21	46,709,983	10,642	4,389
22	50,818,468	20,656	2,460
GL000008.2	209,709	74	2,833
GL000009.2	201,709	80	2,521
GL000194.1	191,469	130	1,472
GL000195.1	182,896	112	1,633
GL000205.2	185,591	202	918
GL000208.1	92,689	15	6,179
GL000213.1	164,239	5	32,847
GL000214.1	137,718	95	1,449
GL000216.2	176,608	388	455
GL000218.1	161,147	105	1,534
GL000219.1	179,198	250	716
GL000220.1	161,802	47	3,442
GL000221.1	155,397	19	8,178
GL000224.1	179,693	86	2,089
GL000225.1	211,173	1,688	125
KI270316.1	1,444	1	1,444
KI270320.1	4,416	2	2,208
KI270330.1	1,652	5	330
KI270337.1	1,121	2	560
KI270429.1	1,361	1	1,361
KI270435.1	92,983	28	3,320
KI270438.1	112,505	223	504
KI270442.1	392,061	682	574
KI270466.1	1,233	4	308
KI270467.1	3,920	27	145
KI270507.1	5,353	2	2,676
KI270510.1	2,415	1	2,415
KI270512.1	22,689	2	11,344
KI270515.1	6,361	1	6,361
KI270519.1	138,126	52	2,656
KI270538.1	91,309	15	6,087
KI270580.1	1,553	1	1,553

KI270581.1	7,046	1	7,046
KI270582.1	6,504	1	6,504
KI270584.1	4,513	2	2,256
KI270587.1	2,969	1	2,969
KI270589.1	44,474	15	2,964
KI270590.1	4,685	8	585
KI270591.1	5,796	9	644
KI270706.1	175,055	97	1,804
KI270707.1	32,032	7	4,576
KI270708.1	127,682	30	4,256
KI270709.1	66,860	64	1,044
KI270710.1	40,176	3	13,392
KI270711.1	42,210	54	781
KI270712.1	176,043	68	2,588
KI270713.1	40,745	112	363
KI270714.1	41,717	25	1,668
KI270716.1	153,799	4	38,449
KI270717.1	40,062	9	4,451
KI270718.1	38,054	17	2,238
KI270719.1	176,845	76	2,326
KI270720.1	39,050	38	1,027
KI270721.1	100,316	5	20,063
KI270722.1	194,050	89	2,180
KI270723.1	38,115	160	238
KI270724.1	39,555	108	366
KI270725.1	172,810	40	4,320
KI270726.1	43,739	22	1,988
KI270727.1	448,248	36	12,451
KI270728.1	1,872,759	185	10,123
KI270729.1	280,839	61	4,603
KI270730.1	112,551	54	2,084
KI270731.1	150,754	17	8,867
KI270732.1	41,543	44	944
KI270733.1	179,772	100	1,797
KI270734.1	165,050	12	13,754
KI270735.1	42,811	29	1,476
KI270736.1	181,920	57	3,191
KI270738.1	99,375	15	6,625
KI270741.1	157,432	5	31,486
KI270742.1	186,739	35	5,335
KI270743.1	210,658	59	3,570
KI270744.1	168,472	138	1,220
KI270745.1	41,891	7	5,984
KI270746.1	66,486	94	707
KI270747.1	198,735	3	66,245
KI270748.1	93,321	3	31,107
KI270749.1	158,759	19	8,355
KI270750.1	148,850	35	4,252
KI270751.1	150,742	50	3,014
KI270753.1	62,944	4	15,736
KI270754.1	40,191	54	744
KI270755.1	36,723	16	2,295
KI270756.1	79,590	22	3,617
KI270757.1	71,251	29	2,456
MT	16,569	50	331
X	156,040,895	23,661	6,594
Y	57,227,415	1,222	46,830
Total	**3,099,061,249**	**814,996**	**3,802**

Number variants by type

Type	Total
SNP	704,083
MNP	0
INS	35,050
DEL	75,863
MIXED	0
INV	0
DUP	0
CNV	0
BND	0
INTERVAL	0
Total	**814,996**

Number of effects by impact

Type (alphabetical order)	Count	Percent
HIGH	5,593	0.139%
LOW	59,405	1.477%
MODERATE	37,552	0.934%
MODIFIER	3,918,485	97.45%

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	36,111	47.102%
NONSENSE	606	0.79%
SILENT	39,948	52.107%

Missense / Silent ratio: 0.904

Number of annotaitons and region counts

Annotation	Region

Type (alphabetical order)	Count	Percent
3_prime_UTR_truncation	1	0%
3_prime_UTR_variant	40,227	0.995%
5_prime_UTR_premature_start_codon_gain_variant	2,058	0.051%
5_prime_UTR_truncation	2	0%
5_prime_UTR_variant	15,753	0.39%
bidirectional_gene_fusion	298	0.007%
conservative_inframe_deletion	321	0.008%
conservative_inframe_insertion	236	0.006%
disruptive_inframe_deletion	723	0.018%
disruptive_inframe_insertion	335	0.008%
downstream_gene_variant	510,906	12.637%
exon_loss_variant	7	0%
frameshift_variant	2,778	0.069%
initiator_codon_variant	5	0%
intergenic_region	268,630	6.645%
intragenic_variant	150	0.004%
intron_variant	2,493,795	61.683%
missense_variant	35,990	0.89%
non_coding_transcript_exon_variant	104,681	2.589%
non_coding_transcript_variant	83	0.002%
splice_acceptor_variant	1,386	0.034%
splice_donor_variant	542	0.013%
splice_region_variant	19,903	0.492%
start_lost	80	0.002%
stop_gained	637	0.016%
stop_lost	80	0.002%
stop_retained_variant	39	0.001%
synonymous_variant	39,915	0.987%
upstream_gene_variant	503,331	12.45%

Type (alphabetical order)	Count	Percent
DOWNSTREAM	510,906	12.706%
EXON	182,780	4.546%
GENE	298	0.007%
INTERGENIC	268,630	6.681%
INTRON	2,476,786	61.596%
SPLICE_SITE_ACCEPTOR	1,250	0.031%
SPLICE_SITE_DONOR	514	0.013%
SPLICE_SITE_REGION	18,358	0.457%
TRANSCRIPT	233	0.006%
UPSTREAM	503,331	12.517%
UTR_3_PRIME	40,198	1%
UTR_5_PRIME	17,751	0.441%

Quality:

Min	30
Max	16,076
Mean	132.053
Median	46
Standard deviation	343.256
Values	30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,9
Count	2769,4534,34733,1519,2370,19990,907,323850,1032,1054,3901,1938,996,2058,2264,2343,2343,872,1279,3056,1427,1335,3222,1470,1888,3290,1515,2038,5822,2866,3799,4385,1960,3282,8

Insertions and deletions length:

Min	0
Max	179
Mean	1.986
Median	1
Standard deviation	7.201
Values	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,6
Count	16454,82353,2425,3076,785,1035,303,710,303,421,151,369,119,190,135,161,86,129,58,85,70,64,44,80,41,34,38,41,24,39,26,27,23,27,21,22,19,16,26,24,15,12,11,19,13,14,17,20,9,18,10,14,13,

Base changes (SNPs)

	A	C	G	T
A	0	19,856	93,756	23,015
C	83,117	0	30,627	100,570
G	102,575	30,743	0	83,016
T	24,270	92,752	19,786	0

Ts/Tv (transitions / transversions)

Note: Only SNPs are used for this statistic.
Note: This Ts/Tv ratio is a 'raw' ratio (ratio of observed events).

Transitions	668,618
Transversions	550,928
Ts/Tv ratio	1.2136

All variants:

```
Sample ,sample1,Total
Transitions ,668618,668618
Transversions ,550928,550928
Ts/Tv ,1.214,1.214
```

Only known variants (i.e. the ones having a non-empty ID field):

```
No results available (empty input?)
```

Allele frequency

Min	50
Max	100
Mean	86.239
Median	100
Standard deviation	22.332
Values	50,100
Count	223869,589521

Allele Count

Min	1
Max	2
Mean	1.725

Median	2
Standard deviation	0.447
Values	1,2
Count	223869,589521

Hom/Het per sample

```
Sample_names , sample1
Reference , 0
Het , 223869
Hom , 589521
Missing , 0
```

Codon changes

How to read this table:
- Rows are reference codons and columns are changed codons. E.g. Row 'AAA' column 'TAA' indicates how many 'AAA' codons have been replaced by 'TAA' codons.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	-	AAA	AAC	AAG	AAT	ACA	ACC	ACG	ACT	AGA	AGC	AGG	AGT	ATA	ATC	ATG	ATT	CAA	CAC	CAG	CAT	CCA	CCC	CCG	CCT	CGA	CGC	CGG	CGT	CTA
-		38	30	52	25	28	142	24	23	20	76	18	13	12	65	37	10	36	53	268	40	20	100	88	92	8	53	43	18	8
AAA	175	1	34	524	21	54			1	205	4			25				45												
AAC	88	89	1	66	820	3	71				186				30	3			36											
AAG	183	541	73	7	34			33		2	1	229				20				72					3					
AAT	94	63	660	22	5				28				291				29				25									
ACA	154	77	1				126	765	85	34			6	144	6	1			2			26								
ACC	157	1	105			156	6	151	679	1	63	4			154								71							
ACG	30			55		1,001	167		115			38	2			395								27						
ACT	109	8	1	6	72	85	589	105	4				77				215								50					
AGA	101	142		1		62				1	35	269	29	36												56				
AGC	109		221				92			50	3	86	598		53												22			
AGG	51			217				23		244	36		44			14												112		
AGT	84			2	270				69	29	466	21					33			1									33	
ATA	86	50			11	180	2	1		16		1			75	96	19													13
ATC	63		22				112	2	7		25		4	159		72	564													
ATG	139		3	35	3		3	436	3			32		142	43	2	98													
ATT	103			1	33				160	4			38	62	432	32						1								
CAA	244	153																	69	572	39	42				171				17
CAC	75		53															43		82	763	1	44				249			
CAG	354		2	156		11												598	94	45	91	6	6	98			1	513		
CAT	85	2			45													40	614	49			2		19				235	
CCA	120					109												99				1	138	939	44	23	1			113
CCC	136						69												68			189	3	219	726		47			
CCG	55							10												58		1,025	171	5	89			63		
CCT	175		1						106												69	57	574	88	3				64	3
CGA	38									52								376				37					32	129	31	9
CGC	59										33								454	1	1		24		1	35	1	53	244	
CGG	59											127								515				63		135	81		45	
CGT	15												24								330				51	24	212	33		
CTA	89										2			62				44				70			1	10				
CTC	46						1						1		42				41				92				48	2		95
CTG	147															83				55			7	287				68	1	550
CTT	202																18	4		2	38				125				17	50
GAA	423	244																81												
GAC	166		244																76											
GAG	274			277																165						1				
GAT	168				156																48									
GCA	113					298																55			4					
GCC	116						384				2	2											43	1						
GCG	61							82												1				29						
GCT	122								305		8														88					
GGA	112									150																24				
GGC	167										353	3															22			
GGG	68			1								201																65		
GGT	109												157																21	
GTA	71												1	179																35
GTC	66														503															
GTG	122															582				4										
GTT	73																256													
TAA	11	8																4												
TAC	103		31																109											
TAG	3																			3										
TAT	90				29																72									
TCA	138			1		54																93								
TCC	143						40		1														161							
TCG	16							7																53						
TCT	100								73																151					
TGA	13									2																9				
TGC	64										15					1											158			
TGG	85											20																256		
TGT	90												7						1										180	
TTA	143													27																103
TTC	68														37		3													
TTG	115															54														
TTT	90																23													

Amino acid changes

How to read this table:
- Rows are reference amino acids and columns are changed amino acids. E.g. Row 'A' column 'E' indicates how many 'A' amino acids have been replaced by 'E' amino acids.
- Red background colors indicate that more changes happened (heat-map).

- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	*	-	?	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
*	35	27			7		4		2			8	7				7	11	2			5	4
-	14		460	287	77	98	171	64	314	93	87	90	285	37	55	300	304	160	307	217	155	19	36
?		1	11																				
A		412		4,401	1	98	199		178							220	1	2	307	1,069	807		
C	45	152	2	1	689			102	82	1			2	1				338	143			61	160
D		334		149		2,260	478		251	124			3		400						77		109
E	129	697		121		434	1,411		287			521	1				246	1			108		
F	4	158			56			963	6		63		333						145		37		63
G	52	456		262	82	237	234	1	2,505			1						486	510		153	58	
H		160				80				1,377		2	39		98	66	214	484					193
I		252						49			1,311	51	151	200	66	1		21	67	464	927		
K	11	358		4			407	1			25	1,073	3	20	162	3	117	436	5	88			
L	43	742		1	6			460		79	149		4,511	137		582	105	146	176	1	436	22	
M		139		3							283	35	147	2	6			32		442	506		
N		182				286			1	61	59	240		3	1,486				477	102			64
P		486		249		1				137			819		1	4,271	157	198	517	294			
Q	74	598		3			197	3		293		309	71		2	152	1,215	685		11			
R	98	323			482				296	785	36	360	151	14		176	892	1,916	201	85		380	
S	55	590		261	192			167	415		86	3	312		491	458	1	241	4,207	336		30	92
T		450		814			1			2	519	147	3	396	179	174		77	311	4,034			
V		332		693	1	59	77	82	105		938		339	582			4		1		1,797		
W	52	85			50				36				80					276	17			2	
Y	65	193			240	44		98		181					60				54				1,284

Variants by chromosome

1, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
1,Count,257,897,713,779,466,378,561,461,437,560,504,625,365,432,281,732,960,715,463,547,482,680,620,563,562,462,495,466,388,416,477,532,411,381,307,284,454,482,336,41

2, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
2,Count,250,398,287,540,189,172,255,246,323,512,701,436,217,111,134,380,238,192,167,251,432,147,89,370,397,431,542,415,445,342,348,293,334,394,141,124,258,359,342,281

3, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
3,Count,154,141,141,250,238,200,112,136,220,356,398,395,290,488,396,256,211,131,73,88,139,126,106,175,141,116,68,186,103,109,118,224,320,213,103,96,120,234,289,190,17

4, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
4,Count,514,588,473,514,308,362,557,632,591,317,260,122,114,127,127,221,190,297,84,53,123,134,148,87,188,325,186,103,105,63,106,111,92,57,63,133,125,200,254,341,308,1

5, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
5,Count,765,904,535,485,299,356,430,375,224,265,494,182,148,270,406,177,344,217,156,154,115,185,94,205,133,136,130,122,114,126,122,353,338,258,313,297,259,253,255,188

6, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
6,Count,411,405,352,485,339,319,361,408,217,156,335,335,177,315,359,372,272,399,313,190,294,331,195,164,350,282,299,356,336,1408,793,1653,2033,874,544,509,563,525,304

7, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
7,Count,997,939,884,399,529,818,623,237,194,149,176,124,250,175,128,129,230,162,136,130,189,329,353,321,194,233,221,268,235,306,320,244,434,211,123,268,360,298,255,26

8, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
8,Count,304,395,258,266,185,193,442,144,250,323,336,450,348,241,98,191,118,321,245,288,216,277,397,360,143,212,219,386,247,223,324,157,91,197,62,82,64,241,322,193,332

9, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,1
9,Count,383,120,198,133,243,169,304,145,128,73,85,134,119,150,245,341,301,224,227,324,167,152,136,100,75,107,145,277,129,108,115,157,289,453,498,346,510,307,327,202,1

10, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
10,Count,511,758,456,692,320,717,648,486,405,160,226,547,655,701,473,414,309,438,342,255,207,275,232,306,329,241,449,506,379,362,463,188,316,269,263,298,204,237,359,4

11, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
11,Count,891,1025,680,497,324,533,376,305,510,444,308,375,362,238,169,205,182,422,520,304,375,126,121,176,150,134,160,151,114,60,116,95,262,282,314,171,187,60,92,134,

12, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
12,Count,583,426,480,544,431,321,615,517,495,462,442,520,342,187,259,128,132,85,107,222,184,311,180,159,188,276,263,311,123,226,377,630,410,142,127,6,12,103,165,126,5

13, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
13,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,413,575,517,415,311,418,558,366,399,349,375,330,386,331,340,285,205,217,268,256,248,298,366,258,261,215,259,338,254,173,1

14, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
14,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,3,0,71,211,370,460,385,293,418,133,83,78,102,69,140,303,141,127,376,220,126,139,100,213,78,86,112,61,94,118,86,112,84,238,365

15, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
15,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,20,6,51,548,124,474,322,234,413,258,194,238,254,176,249,169,301,242,121,133,93,124,182,493,475,417,225,229,316,71,103,168,1

16, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
16,Count,1009,1031,551,619,720,537,311,295,575,334,544,669,434,292,280,417,292,300,209,486,327,297,253,428,341,326,203,444,451,463,480,416,104,119,190,149,33,4,12,0,0

17, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
17,Count,469,609,502,605,703,460,385,662,496,441,362,242,198,189,176,310,262,333,442,249,253,929,208,4,3,14,106,303,251,318,358,295,277,195,272,340,356,226,343,473,36

18, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
18,Count,574,189,308,400,172,203,293,319,351,398,384,379,469,384,362,82,0,5,9,1,5,287,261,350,334,195,245,135,92,108,125,331,208,110,137,234,299,290,131,104,134,114,1

19, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
19,Count,893,1043,925,934,881,677,674,914,1083,599,720,658,512,537,840,758,653,847,727,522,577,526,423,560,59,0,0,127,277,410,253,185,509,633,416,656,546,411,739,772,

20, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
20,Count,521,378,436,547,418,490,215,164,191,231,250,156,108,302,165,151,239,402,417,262,282,199,175,431,296,435,80,1,206,1132,527,353,488,458,432,452,395,462,399,200

21, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
21,Count,0,0,0,0,0,87,29,56,146,368,809,0,6,259,268,149,126,224,149,137,165,153,131,173,138,281,207,156,231,218,201,412,385,435,250,123,369,314,190,300,200,437,581,34

22, Position,0,1000000,2000000,3000000,4000000,5000000,6000000,7000000,8000000,9000000,10000000,11000000,12000000,13000000,14000000,15000000,16000000,
22,Count,0,0,0,0,0,0,0,0,0,0,286,552,253,0,0,77,300,486,208,513,565,326,611,850,472,517,406,396,226,542,503,447,438,324,272,318,684,834,736,629,387,685,773,719,847,70

GL000008.2, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
GL000008.2,Count,0,15,9,7,3,4,1,9,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

GL000218.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
GL000218.1,Count,2,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2,0,0,0,0,0,1,0,3,2,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,4,9,11,6,3,0,1,1,0,1,0,0,0,0,4,0,0,0,5,4,2,0,3,1,0,0,2,0,0,0,0,0,0,0,1,

GL000219.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
GL000219.1,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,5,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2,0,4,13,5,2,8,0,1,14,0,1,0,0,0,3,5,0,0,0,0,2,0,0,0,0,0,0,0,1,4,0

GL000220.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
GL000220.1,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

KI270515.1, Position,0,100,200,300,400,500,600,700,800,900,1000,1100,1200,1300,1400,1500,1600,1700,1800,1900,2000,2100,2200,2300,2400,2500,2600,2700,2
KI270515.1,Count,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

KI270714.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270714.1,Count,0,0,0,0,0,2,1,0,0,0,0,2,0,1,0,0,0,0,0,0,0,0,4,0,0,1,2,0,0,0,1,1,1,1,2,0,0,0,1,3,2,0

KI270716.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270716.1,Count,0,0,1,0,0,0,0,0,0,0,0

KI270733.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270733.1,Count,3,0,0,0,0,0,0,0,3,0,0,3,6,1,0,0,1,0,0,0,0,0,0,0,0,2,0,1,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,1,0,2,0,0,0,0,0,0,0,0,0,0,0,0

KI270734.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270734.1,Count,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,2,0,0,0,0

KI270735.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270735.1,Count,3,0,0,0,0,0,0,0,0,1,2,0,0,0,0,1,0,3,0,0,0,0,0,0,0,0,0,2,0,0,0,0,0,0,0,0,2,0,0,0,7,8

KI270736.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270736.1,Count,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

KI270756.1, Position,0,1000,2000,3000,4000,5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000,18000,19000,20000,21000,22000,2300
KI270756.1,Count,4,1,7,0,10,0,0,0